# Georgios Mavromatidis | Resumé

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## Education

**ETH Zürich** **Switzerland**
*PhD in Urban Energy Systems* *2013–2017*

*Dissertation Title:* "Model-based Design of Distributed Urban Energy Systems under Uncertainty" *Supervised by:* Prof. Dr. Jan Carmeliet and Dr. Kristina Orehounig

*Parallel activities: Zernez Energia 2020 project* – Main tasks: Technoeconomic modelling of a Swiss village's energy system and scenario analysis for the achievement of a carbon neutral energy supply utilizing only local, renewable energy sources. More info at: `http://www.zernezenergia2020.ch`

**Imperial College London** **UK**
*MSc in Sustainable Energy Futures* *2011–2012*

*Overall Grade: Distinction*, Awarded Prize for best overall student of the year from a cohort of 40 students

*MSc Thesis:* "Diagnostic tools of energy performance for supermarkets using Artificial Neural Network algorithms" *Supervised by:* Prof. Dr. Nilay Shah and Dr. Salvador Acha

**Aristotle University of Thessaloniki** **Greece**
*MEng in Mechanical Engineering* *2005–2010*

*Overall Grade: Excellent (8.94/10*, 1st place of success among a cohort of 120 students)

*MEng Thesis:* "Investigation of the operational impact of Phase Change Materials (PCMs) in domestic refrigerators" *Supervised by:* Prof. Dr. Agis M. Papadopoulos

## Experience

**Chair of Building Physics, ETH Zürich** **Switzerland**
*Postdoctoral Researcher* *2017–present*

Activities involved include:
- Participation in Phase II of the Swiss Competence Center for Energy Research on Future Energy Efficient Buildings & Districts (SCCER FEEB&D).
- Teaching in "Building Physics", "Building Systems" and "Whole Building Simulation" courses.
- Google Summer of Code Mentor | Empa Laboratory for Urban Energy Systems - June - August 2017

**Centre for Process Systems Engineering, Department of Chemical Engineering, Imperial College London** **UK**
*Research Assistant* *2012-2013*

Projects involved include:
- Imperial College – Sainbury's partnership which aims at reducing the carbon footprint of supermarkets by implementing novel energy strategies that respond to the effects of climate change and increasing energy prices.
- Development of the agent-based human ecosystem model ARES Prototype (**A**gent–Based **R**esource **E**conomic **S**imulator)

**Laboratory of Heat Transfer and Environmental Engineering (LHTEE), Department of Mechanical Engineering, Aristotle University Thessaloniki** **Greece**
*Research Assistant* *2010–2011*

Main research projects carried out include:
- Evaluation of night time electricity tariffs for refrigeration purposes
- Thermodynamic analysis of a solar operated domestic absorption refrigerator
- Assessment of energy conservation measures in typical buildings according to the geographic and climatic regions

## Awards/Distinctions

- Imperial College's Energy Futures Lab Award for best overall student of the year 2011/2012 from a cohort of 40 students

- State Scholarships Foundation Award (IKY) Honorary Scholarships for distinction in my studies during the academic years 2005/06 & 2006/07

## Computer skills

**Programming**: Matlab, R, Python, gPROMS, NetLogo
**Optimization**: AIMMS, GAMS
**Energy**: EnergyPlus, EnergyPRO, RETScreen
**GIS**: ArcGIS, QGIS

## Languages

**Greek**: Native Speaker
**English**: **Professional Proficiency**: TOEFL iBT Score: 115/120
**German**: **Basic**: B1

## Journal Publications

1. **Mavromatidis G.**, Orehounig K., Carmeliet J. *Optimal design of distributed energy systems under uncertainty: A two-stage stochastic programming approach.* Applied Energy, submitted for publication.
2. **Mavromatidis G.**, Orehounig K., Carmeliet J. *Uncertainty and global sensitivity analysis for the design of distributed energy systems.* Applied Energy, under review.
3. **Mavromatidis G.**, Orehounig K., Carmeliet J. *A review of uncertainty characterisation approaches for the optimal design of distributed energy systems.* Renewable and Sustainable Energy Reviews, revised version submitted.
4. Wu, R., **Mavromatidis G.**, Orehounig K., Carmeliet J. *Multiobjective Optimisation of Energy Systems and Building Envelope Retrofit in a Residential Community.* Applied Energy, 190, 634-649.
5. **Mavromatidis G.**, Orehounig K., Richner P., Carmeliet J. (2015) *A strategy for reducing $CO_2$ emissions from buildings with the Kaya identity - A Swiss energy system analysis and a case study.* Energy Policy, 88, 343-354.
6. Allegrini J., Orehounig K., **Mavromatidis G.**, Ruesch F., Dorer V., Evins R. (2015) *A review of modelling approaches and tools for the simulation of district-scale energy systems.* Renewable and Sustainable Energy Reviews, 52, 1391-1404.
7. **Mavromatidis G.**, Orehounig K., Carmeliet J. (2015) *Evaluation of photovoltaic integration potential in a village.* Solar Energy, 121, 152-168.
8. Orehounig K., **Mavromatidis G.**, Evins R., Dorer V., Carmeliet J. (2014) *Towards an energy sustainable community: an energy system analysis for a village in Switzerland.* Energy and Buildings, 84, 277-286.
9. **Mavromatidis G.**, Acha S., Shah N. (2013) *Diagnostic tools of energy performance for supermarkets using Artificial Neural Networks.* Energy and Buildings, 62, 304-314.

## Conference Proceedings

1. **Mavromatidis G.**, Orehounig K., Carmeliet J. (2017) *Designing electrically self-suffcient distributed energy systems under energy demand and solar radiation uncertainty.* International Conference Future Buildings & Districts (CISBAT 2017), Lausanne, Switzerland, September 6-8.
2. Marquant J.F., **Mavromatidis G.**, Evins R., Carmeliet J. (2017) *Comparing different temporal dimension representations in distributed energy system design models.* International Conference Future Buildings & Districts (CISBAT 2017), Lausanne, Switzerland, September 6-8.
3. Orehounig K., Landolt J., **Mavromatidis G.**, Omu A., Wang D., Wu R. (2017) *A Method to Support Multi-Criteria Decision Making for Building Systems Update at Urban Scale.* 18th International Conference of the International Building Performance Simulation Association (BS 2017), San Francisco, CA, USA, August 7-9.
4. **Mavromatidis G.**, Orehounig K., Carmeliet J. (2017) *Trade-offs between risk-neutral and risk-averse decision making for the design of distributed energy systems under uncertainty.* 30th International Conference on Efficiency, Cost, Optimization, Simulation and Environmental Impact of Energy Systems (ECOS 2017), San Diego, CA, USA, July 2-7.
5. Wu, R., **Mavromatidis G.**, Orehounig K. (2016) *Reliability Optimisation of a District Multi-Energy System.* 19. Status-Seminar, ETH Zurich, Switzerland, September 8-9.
6. **Mavromatidis G.**, Orehounig K., Carmeliet J. (2016) *Uncertainty and sensitivity analysis for the optimal design of distributed urban energy systems.* Sustainable Built Environment Conference (SBE16), Zurich, Switzerland, June 13-17.
7. Wu, R., **Mavromatidis G.**, Orehounig K., Carmeliet J. (2016) *Optimal Energy System Transformation of a*

*Neighbourhood.* Sustainable Built Environment Conference (SBE16), Zurich, Switzerland, June 13-17.

8. **Mavromatidis G.**, Orehounig K., Carmeliet J. (2015) *Evaluation Of Solar Energy Integration Potential In A Neighborhood.* 14th International Conference of the International Building Performance Simulation Association (BS 2015), Hyderabad, India, December 7-9.
9. **Mavromatidis G.**, Orehounig K., Carmeliet J. (2015) *Climate change impact on the design of urban energy systems.* International Conference Future Buildings & Districts (CISBAT 2015), Lausanne, Switzerland, September 9-11.
10. Orehounig K., **Mavromatidis G.**, Derome, D., Carmeliet J. (2015) *'Photovoltaic Panels as a Main Component of Energy Sustainable Communities: Comparative Energy Analysis of a Village Under Swiss and South African Climatic Loads'.* Third Southern African Solar Energy Conference (SASEC 2015), Skukuza, Kruger National Park, South Africa, May 11-13.
11. Orehounig K., **Mavromatidis G.**, Evins R., Dorer V., Carmeliet J. (2014) *Predicting energy consumption of a neighborhood using building performance simulation.* Building Simulation and Optimization conference (BSO 2014), London, United Kingdom, June 23-24.
12. **Mavromatidis G.**, Evins R., Orehounig K., Dorer V., Carmeliet J. (2014) *Multi-objective optimization to simultaneously address energy hub layout, sizing and scheduling using a linear formulation.* 4th International Conference on Engineering Optimization (ENGOPT 2014), Lisbon, Portugal, September 8-11.
13. Acha S., **Mavromatidis G.**, Caritte V., Shah N. (2013) *Effective Low-cost Energy Saving Strategies in Supermarkets: A UK Case Study.* 26th International Conference on Efficiency, Cost, Optimization, Simulation and Environmental Impact of Energy Systems (ECOS 2013), Guilin, China, July 16-19.
14. Acha S., **Mavromatidis G.**, Caritte V., Shah N. (2013) *Techno-economical Technology Assessment for Operational Zero Carbon Supermarkets.* 26th International Conference on Efficiency, Cost, Optimization, Simulation and Environmental Impact of Energy Systems (ECOS 2013), Guilin, China, July 16-19.
15. **Mavromatidis G.** & Papadopoulos A. M. (2011) *Thermodynamic analysis of a solar operated domestic absorption refrigerator.* Proceedings of the National Conference on Architecture, Energy and Environment in buildings and cities (ARENEP 2011) (A.M. Papadopoulos, ed), Athens, Greece, 3-4 May (in Greek).

## Invited talks/Presentations

**Jun 2017**: Invited presentation given as part of the Empa RFA Colloquium Natural Resources and Pollutants + Energy, titled *"How can we design effective urban energy systems in an uncertain world?"*, hosted by Empa

**Nov 2016**: Invited presentation given as part of the Empa's Topical Day on "High-performance multiscale modelling III" titled *"Uncertainty and sensitivity analysis for the design of distributed urban energy systems"*, hosted by Empa

**May 2015**: Invited presentation given as part of the HSLU BSc program in Energy Systems Engineering titled *"Zernez Energia 2020: Designing an energy sustainable community"*, hosted by Hochschule Luzern

**Dec 2014**: Invited presentation given as part of the Empa RFA Colloquium Natural Resources and Pollutants + Energy, titled *"Energy hub modelling for the optimal integration of solar energy in the built environment"*, hosted by Empa

**Oct 2014**: Invited presentation given as part of the Empa Ph.D. + Research Focal Area Seminar, titled *"Zernez Energia 2020 - Designing an energy sustainable community"*, hosted by Empa

**Feb 2013**: Invited lecture given to a graduate class on *Principles of supermarket energy systems*, hosted by Imperial College London

**Feb 2013**: Invited lecture given to a graduate class on *Artificial Neural Networks for energy monitoring and analysis*, hosted by Imperial College London

**Nov 2012**: Presentation given at *Climate KIC Professional Education Event*, hosted by Imperial College London

## Student supervision

- Title: *Emulation of energy optimisation models using Machine Learning algorithms*, (MSc thesis, ETH Zurich)
  Student: Emmanouil Thrampoulidis
- Title: *Meta-model based Sensitivity Analysis for the Design of Urban Energy Systems*, (Sem. thesis, ETH Zurich)
  Student: Emmanouil Thrampoulidis
- Title: *Reliability Optimisation of District Multi-Energy Systems*, (MSc thesis, ETH Zurich)
  Student: Raphael Wu
- Title: *Optimal Energy System Transformation of Residential Buildings*, (Sem. thesis, ETH Zurich)
  Student: Raphael Wu
- Title: *Technoeconomic Assessment of CHP Installation in AHEPA Hospital*, (Diploma thesis, Aristotle University)
  Student: Angelos Sotiriou